Marion CHAUVEAU

PERSONAL INFORMATION

Date of Birth: September 19, 1997
Nationality: French
Email: chauveaumarion4@gmail.com
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EDUCATION

2022 – 2025
PhD in Biological Physics, TIMC laboratory (Grenoble-Alpes University) & Gulliver laboratory (ESPCI Paris), France.
Generative models for protein sequences. Advisors: Ivan Junier and Olivier Rivoire. [pdf]
Main projects: Overcome undersampling-induced biases in generative models for protein sequences · Investigate protein properties with statistical models.

2021 – 2022
MSc in Applied Mathematics, Mathématiques / Vision / Apprentissage, ENS Paris-Saclay, France
Coursework: Probabilistic graphical models · Introduction to Digital Imaging · Computational statistics · Convex optimization and applications in machine learning · Generative Models for Images · Graphs in Machine learning · Algorithms for speech and natural language processing.

2020 – 2021
MSc in Physics, Soft Matter and Biological Physics, ENS-ICFP, Paris, France.
Coursework: Statistical Physics · Biophysics · Plasma Physics and advanced fluid dynamics · Soft matter physics · Physics of fluids · Information, inference, networks: from statistical physics to big biological data · Complex systems: from physics to social sciences · Statistical physics: Disordered systems and interdisciplinary applications · Brownian Motion and Stochastic Process.

2018 – 2019
BSc in Physics, Fundamental physics, ENS Paris-Saclay, France.

2015 – 2018
CPGE PCSI-PSI*, Lycée Clemenceau, Nantes, France.

RESEARCH

October 2025 – January 2026
Postdoctoral researcher, Gulliver laboratory (ESPCI Paris), France.
Generative models for protein sequences. Advisor: Olivier Rivoire.

2022 – 2025
PhD in Biological Physics, TIMC laboratory (Grenoble-Alpes University) & Gulliver laboratory (ESPCI Paris), France.
Generative models for protein sequences. Advisors: Ivan Junier and Olivier Rivoire.

April 2022 – September 2025
Research internship, Centre Borelli, ENS Paris-Saclay, Université Paris-Saclay, France.
Human motion analysis via graph signal dictionary learning. Advisor: Laurent Oudre.

April 2021 – July 2021
Research internship, Centre interdisciplinaire de recherche en biologie (CIRB), Collège de France, Paris, France.
Generative models for bacterial genome organization. Advisor: Olivier Rivoire.

November 2019 – December 2019
Research internship, Laboratoire de Physique des Plasmas (LPP), École Polytechnique, Palaiseau, France.
Study of density and potential fluctuations in a magnetised toric plasma. Advisor: Pierre Morel.

May 2019 – June 2019
Research internship, Research Laboratory in Hydrodynamics, Energy and Atmospheric Environment (LHEEA), Centrale Nantes, France.
Study of the performances of the small swell water tank beach. Advisor: Félicien Bonnefoy.

PUBLICATIONS

Under review

- **M. Chauveau**, Y. Kleeorin, E. Hinds, I. Junier, R. Ranganathan, O. Rivoire. Improved inference of multiscale sequence statistics in generative protein models. In review, Plos CB journal. [bioRxiv]
- P. Guénon, **M. Chauveau**, K. Filipowska, I. Tunon, C. Nizak, K. Reynolds, G. Stirnemann, O. Rivoire, D. Laage. Mechanism of catalytic regulation by an allosteric hot spot in dihydrofolate reductase. [chemRxiv]
- M. Jullien, **M. Chauveau**, S. Chobert, E. Bouvet, W. Schmitt, F. Pierrel, S. Abby, N. Varoquaux, I. Junier. COCOA-Tree: Phylogenetic visualization and comparative analysis of coevolving residues. In review, PCI-MCB journal. [bioRxiv]

Peer-reviewed

- S. W. Combettes, P. Boniol, A. Mazarguil, D. Wang, D. Vaquero-Ramos, **M. Chauveau**, L. Oudre, N. Vayatis, P.-P. Vidal, A. Roren, and M.-M. Lefèvre-Colau. Arm-CODA: A Data Set of Upper-limb Human Movement During Routine Examination. In Image Processing On Line, 14:1-13, 2024. [pdf]
- **M. Chauveau**, A. Mazarguil, L. Oudre. Graph dictionary learning for the study of human motion. In EMBC 2024, pp. 1–5. [pdf]

TALKS

- Predicting compensatory mutations in proteins using statistical models - BiGre Days, February 2025. [slides]
- Graph dictionary learning for the study of human motion - Conference of the IEEE EMBS, Summer 2024. [slides]
- Generative models of protein sequences - Journées de la Physique Statistique, Winter 2024. [slides]
- Generative models of Protein sequences - 5th course on Multi-scale Integration in Biological Systems, Fall 2023. [slides]

TEACHING

2023 – 2024
Oral examination in Physics and Chemistry (CPGE “Khôlle”) (25 h), Lycée Descartes, Tours, France

September 2023 – December 2023
Practical works in Physics for L1 students (20 h), Sorbonne University, Paris, France

September 2023 – December 2023
Numerical projects in Physics for L1 students (10 h), Sorbonne University, Paris, France

November 2019 – March 2020
Practical works in Physics for high-school students, Lycée de Cachan, France

LANGUAGES

French - Native
English - Fluent, Cambridge Certificate in Advanced English (level C1)